

People keep the moment and lose the date, but Photos indexes photos, not moments

Strategic Goal (Brief p. 2)

"Increase the percentage of users who successfully retrieve a photo they remember but cannot precisely describe."

Core Finding Across Two Independent Research Methods

77.1% of observed search failures occur before recovery can help: the query clue is misread (35.4%) or the photo never surfaces in results (41.7%). (§B)

Survey (n=15) and user testing (n=6) agree: people keep *roughly when* and the event, but lose the calendar date. 4 of 5 real-app users strip time into bare nouns, and none of the 5 found their photo. (§G1, §G2)

85,140

Public posts analyzed across 4 platforms in discovery engine

+0.075

Recall@20 gain weighted by real memory (0.259 → 0.334); moment@5 +0.111

77.1%

Failures at interpretation & surfacing before recovery

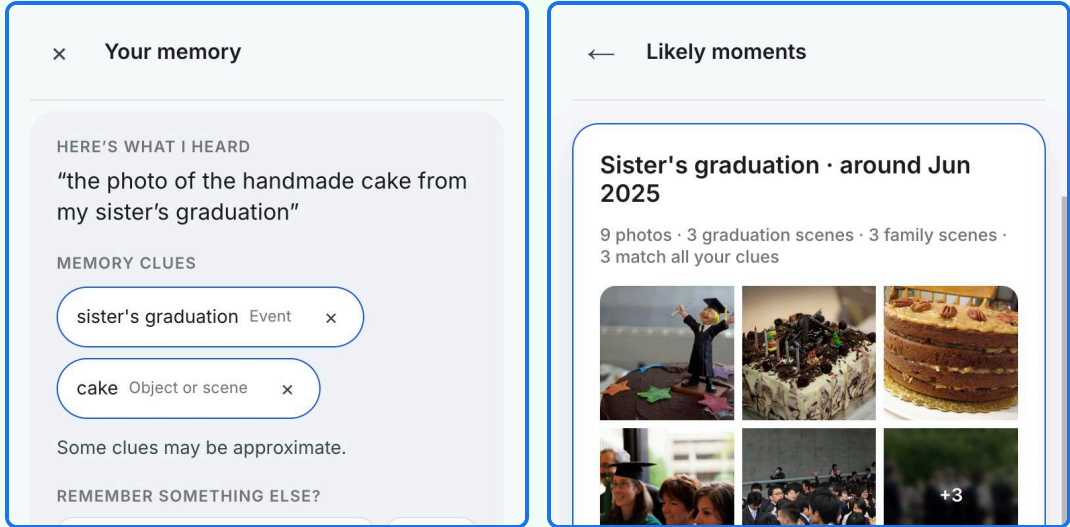
4.0 / 5

Ease vs Google Photos, from the 5 testers who use it

What Was Built & Deployed Live

1. Discovery Engine: Structured pipeline turning 85,140 posts into 144 specific attempts with blind two-model audit.

2. Memory Trails MVP: Episode-first re-entry prototype inside Google Photos. Resolves vague dates, clusters moments, and explains matches with soft scoring.



retrieval-discovery-engine.vercel.app

memory-trails-v2.vercel.app

Retrieval breaks most at interpretation and surfacing, and almost never at recovery

North Star: URR (User Retrieval Rate)

Share of users with a vague-memory search in 28 days who reach the photo. A/B splits by user.

$$URR = Expression \times [1 - (1 - Interpretation \times Surfacing \times Recognition)^{\bar{n}}]$$

Product Outcome	The Brief's Guiding Question	Engine (144)	Survey (15)	What People Report
Expression	Can't express what they remember?	2 (1.4%)	3 (20%)	Don't know what to type; 2 of 3 never search
Interpretation ★	Does Photos misread the clues?	51 (35.4%)	3 (20%)	Add a year, reword: "You must now specify the year"
Surfacing ★	Is the photo in results at all? (our term)	60 (41.7%)	5 (33%)	A grid of near-misses; target absent
Recognition	Are the results hard to judge?	2 (1.4%)	2 (13%)	Can't confirm; 7 of 15 end unsure
Recovery	Can't refine a failed search?	1 (0.7%)	1 (7%)	2–3 tries for 9 of 13 → $\bar{n}$

Strategic Metric Takeaways

- 1. Where Google invested: Ask Photos and the hybrid router read prompts better, but don't group moments or resolve event-relative time.
- 2. 77% in two stages: interpretation + surfacing = 111 of 144 failures.
- 3. Expression disagrees (1.4% vs 3 of 15), but 2 of those 3 browse instead; the moment view serves them.

Browse path (outside the formula)

Survey: 10 of 15 scrolled or opened an album first. MVP testers last found old photos by scroll 3, album 2, search 0.

Recovery scoped out

0.7% of engine failures, 1 of 15 in survey. Kept as a safety net, tracked as a diagnostic.

The engine turns 85,140 public posts into 144 retrieval attempts, each recording what was remembered, what was forgotten, and where search broke

1 · COLLECT

85,140

Public posts
(Play Store,
App Store, YT,
Reddit)

2 · SCREEN

819

Relevant (Gate
A 5,305 → Gate
B 1,333
sample)

3 · EXTRACT

144

Specific
attempts (720
episodes; 62
scoreable)

4 · AUDIT

203

Pairs checked
blind by Qwen
27B model

5 · COMPARE

9 Areas

Ranked only on
agreed fields
(O1–O9)

Extracted Field in Schema	The Brief's Underlying Question Answered	Reliability Check
asset_type	What kinds of old photos do users struggle to retrieve?	Validated in Gate B
cues_retained (14 cue types)	What do people actually remember about the photo?	Jaccard 0.557
cues_lost (date, place, album, words)	What have they forgotten when search fails?	Jaccard 0.673
query_verbatim · search_mode	How do they search when memory is incomplete?	Quote verify 85.2%
failure_stage (5 stages)	Where does retrieval break? (Part 3 decomposition)	κ 0.509 (Moderate)

Why This Is More Than Summarisation

1. Structure, not sentiment: Every review was converted into a structured, typed record with fixed vocabularies, not qualitative impression tags.
2. Pre-registered rules: Hypotheses H1–H5 and decision thresholds were frozen before data collection, preventing post-hoc confirmation bias.
3. The audit overruled the ranking: When the blind audit showed hypothesis agreement was low (Jaccard 0.347) while failure stage agreement was solid (κ 0.509), the engine demoted hypothesis ranking and pivoted to failure stages.

retrieval-discovery-engine.vercel.app

People remember roughly when, but search needs the exact date, and the exact date is what they forget

The Brief's Four Questions Answered (Engine vs. Survey)

Question	Engine (144 Specific Attempts)	Survey (n=15 / 14 Dedup)
Photo Kinds	Personal photos 80 · multi-photo 26 · videos 16 · screenshots 7 · docs 3	All 3 real-trouble cases were document or medicine photos
Remembered	Roughly when 40 · object 17 · exact date 14 · text 12 · people 10	Roughly when 8 · object 7 · people with me 6
Forgotten	The date 37 · album 29 · exact words 10 · place 10	When it was taken 9 · words to search 9
Search Method	20 of 32 queries are 1 word ("dog"); 10 name a time ("Halloween 2024")	First move: scroll 9/15; own-app: 4 of 5 strip time to bare nouns

Opportunity Areas: Why O1 Leads (§B2)

O1 (Rough time or an event): 42 attempts, 20 not surfaced. *Core focus.*

O8 (No clue at all): 47 attempts. Larger, but users kept zero search clues. O1 is the largest group where user memory exists and search fails to use it.

Excluded: O7 (Exact date known, 14 — regular search works); O9 (Path moved by UI update, 12 — app design).

Pre-Registered Hypothesis Verdicts (Rule B)

H1 Episodic time: SUPPORTED (root cause).
Roughly when is the #1 kept cue (40); the date is the #1 lost cue (37).

H2 Recognition: SUPPORTED (secondary). 41.7% never surfaced; 7 of 15 ended unsure.

H3 Dead-end recovery: NOT THE LEAD.
Overruled by the audit: recovery is 0.7% of failures (stage κ 0.509).

H4 Hinglish: UNTESTED. Weak signal (2 of 15 survey, 1 of 6 tests); kept as a design choice.

H5 Text not indexed: LOW VOLUME (3 docs), but high harm.

H6 Path changed: MINOR (8.3%), found post hoc.

Most people scroll before they search, and nearly half end close but unsure

Research Methodology Rationale

Structured interviews by form (n=15, 23–30 Sep): every answer maps onto the engine's vocabulary.

Why: 100% scoreable stories (vs 8.6% of public posts); reached 15 people vs 2 call contacts.

Costs: self-reported, convenience sample; 8 of 15 fit the segment.

[Survey questionnaire \(Google Form\) ↗](#)

Survey vs. Engine (Side-by-Side)

Metric	Engine (144)	Survey (15 / 14)
Most remembered	Roughly when (40)	Roughly when (8)
Most forgotten	The date (37)	When taken (9)
Top failure	Never surfaced (41.7%)	Never surfaced (5 / 4)
Ended unsure	Not measurable	7 of 15 (47%)
First move: scroll	15 workarounds	9 of 15 (60%)

Reported Retrieval Incidents (Survey Respondents)

ID	Target Photo	What They Remembered	What They Typed	Break	Outcome
S04	Wedding photo	Roughly when, event, people	"wedding, pichle saal diwali"	Misread	Unsure
S02	Personal photo	Roughly when, event, object	"gym"	Eval fail	Unsure
S03	Personal photo	Roughly when, event	(Didn't know what to type; scrolled)	Expression	Found
S11	Personal photo	Roughly when, event	(Never typed; opened album)	Expression	Unsure
S06	Medicine bill	Object, text in bill	"medicine, bill"	Surfacing	Never found

"wedding, pichle saal diwali" — S04 (Hinglish: a festival and relative year, not a calendar date)

"medicine, bill" — S06 (Never found; had to get the document again)

Ask Photos Reality: 7 of 15 had never heard of it. All 4 who described a result said it showed "related photos, but not the one I wanted" and that they "could not tell why".

The segment is people who can place the moment only roughly, and search can't turn "roughly" into a date window

Chosen Target Segment

Long-time Google Photos users searching for a personal moment they can place only roughly.

Defined by **cognitive memory state** (what they remember), not demographics: "around Diwali", "my sister's graduation", "last winter".

Segment Sizing Filter	Engine (144)	Survey (15)
Kept rough time or event (O1)	42 (29%)	8 (53%)
Target Segment (After Exclusions)	37 (26%)	8 (53%)
Personal photo or video	30 of 37	All 8
Large library (5,000+ items)	Not measured	5 of 8

Deliberate Exclusions: 14 who remembered exact date (standard search works) · 12 whose path moved in an update (H6 navigation) · Cloud sync / corruption issues.

The Root Cause: The Broken Link

Human Memory Stores
EPISODES

Rough time, event anchors, surrounding scenes ("last winter", "cousin's wedding")



Google Photos Indexes
ITEMS & DATES

Isolated photos with calendar timestamps (YYYY-MM-DD) and tags

"I'll type in something super simple, like 'Halloween 2024' and it seriously can't find anything?" — Play Store review

How Retrieval Breaks for the Segment (37 Attempts):

- **Misinterpretation (12):** Query clue is misread; search fails to map the event to a date window.
- **Surfacing Failure (18):** Returns an unranked grid across years; target photo never surfaces.
- **Real-App Probe Verification (§G2):** 4 of 5 users compressed memory into bare nouns ("vacation", "gym"), and 0 of 5 found the photo.

People already scroll to "roughly when" by hand and still end unsure, so the product should take that step for them

PROBLEM DEFINITION

"People remember when-ish and what happened; Google Photos indexes items and calendar dates. So the photo is in the library, the person can describe the moment but not the photo, and it never comes back."

The Workaround is the Design Blueprint

Engine: 15 of 21 workarounds are manual scrolling (7 of 10 failed): *"Every time I scrolled to the right time frame, it kept loading more pictures..."*

Survey: Scrolling was the first move for 9 of 15. 5 of those 9 still ended unsure.

MVP testers: last found old photos by scroll 3, album 2, search 0.

Insight: people already retrieve by moment, by hand.
The product should find the time window and group the moments.

Solution Rationale: Why This, Not Ask Photos

AI only where it's needed: (1) turning clues into date spans; (2) grouping photos into recognisable moments. Uses existing metadata.

Why not Ask Photos: no editable clues, no moment grouping, no "why this". **Everyone who tried it got "related photos, but not the one": 4 of 4 in survey, 3 of 3 in tests.**

Scoped out: clarifying question (1.4%) · full recovery agent (0.7%), replaced by 1-tap clue alternatives.

HOW THE THINKING EVOLVED

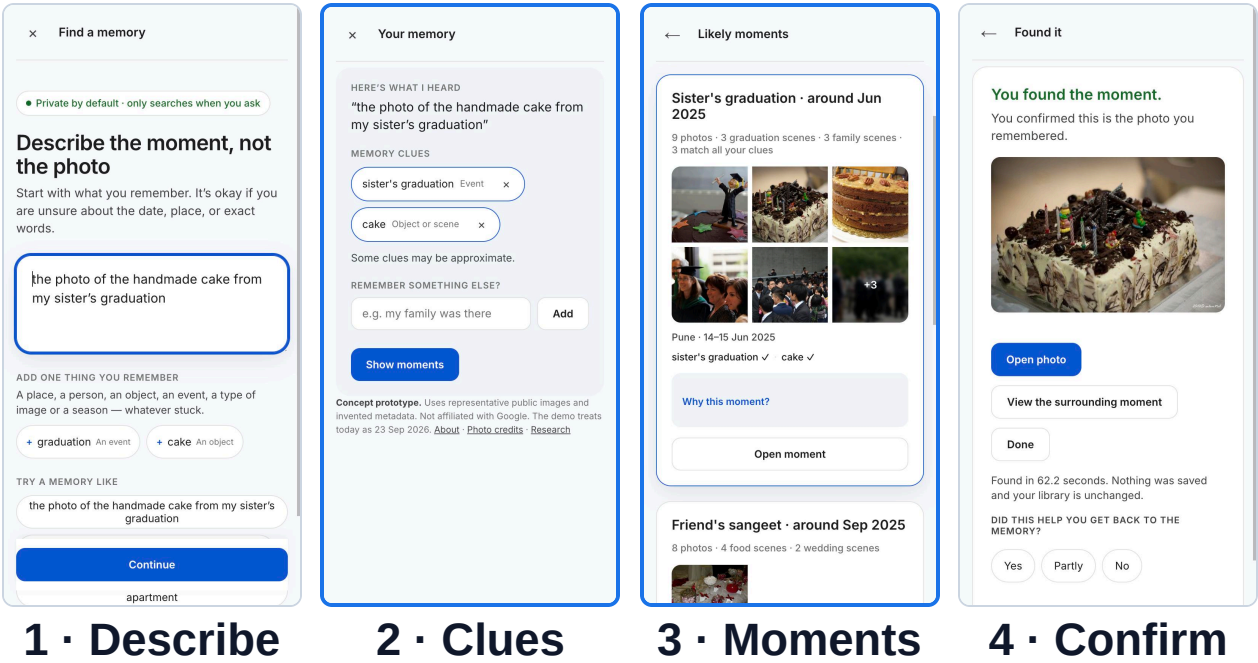
1. Metric: per-search success → per-user URR	2. Outcomes: recovery agent → 5 stages + browse	3. Discovery: H3 ranked top → overturned by audit	4. Reported behaviour: retry search → scroll by hand	5. Problem: "search is bad" → items vs. moments
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Memory Trails turns "roughly when and where" into moments you can recognise

Placement & flow

Where it lives: inside Photos search. When a search fails, *"Can't describe it?"* opens it with the words carried over.

Soft scoring: a date outside the window lowers rank instead of hiding the photo.



memory-trails-v2.vercel.app ↗

Blue frame = AI step

Empirical Evaluation Ladder (120 Tasks, 1,282 Photos)

Cue Dropout Level	What the Query Keeps	Plain CLIP	Memory Trails	Moment@5
L3 (Fully described)	Vague time + exact place + library word	0.172	0.962	0.933
L2 (Two vague cues)	Paraphrased content + 2 vague cues	0.209	0.276	0.333 ★
L1 (One vague cue)	Paraphrased content + 1 vague time cue	0.242	0.309	0.333
L0 (Content only)	Pure scene description, no metadata	0.312	0.312	0.267
Weighted by real memory	Weighted by real cue counts (§E1b)	0.259	0.334 (+0.075)	0.336 (+0.111)

Read this first: L3's 0.962 needs three cues; only 4% (6 of 144) of real attempts kept that many. Weighted by real memory, recall gains **+0.075** and moment@5 **+0.111**: grouping into moments helps more than flat ranking.

"This MVP validates the memory-reentry interaction and recovery model using representative media. It does not validate production-scale Google Photos retrieval accuracy."

5 of 6 testers were sure they found the photo and 4 of 5 found it easier than Google Photos, but compound dates still trip it up

ID	Own Photos search → result	Task 1 (cat)	Task 2 (dog)	Sure	vs GP	Would use
R01	"Wedding before COVID" → wrong years	Found, sure	Not found	5/5	5/5	Every time
R02	"gym" → can't tell which	Found, sure	Found (1-tap alt)	4/5	4/5	When fails
R03	(Apple Photos user)	Found, sure	Found (nearby)	5/5	—	Every time
R04	"vacation" → wrong years	Found, sure	Found (1-tap alt)	4/5	4/5	When fails
R05	"Wedding event" → can't tell which	Found, sure	Found (clue edit)	4/5	4/5	Every time
R06	"Kerala photos" → can't tell which	Right day, unsure	Found (1-tap alt)	4/5	3/5	Every time

6 of 6

Fit the segment

4.33 / 5

Task 1 sureness

4.0 / 5

Ease vs Photos

0 of 5

Found it in own Photos

MVP test form ↗

memory-trails-v2.vercel.app ↗

What testers said

"Shows the moment, not a grid" — R02

"Moving to nearby months, like scrolling but faster" — R03

"Works without knowing what to type" — R04

"Finds exact month & year... understands mix of Hindi and English" — R06

What broke → next iteration

1. Compound time: R01 stuck on "pichle ke pichle saal" (year before last). Fix: parse compound offsets; editable year.

2. Clues missed: R04 didn't notice them. Fix: high-contrast "Showing Diwali 2025 · Tap to change".

3. Speed: R05, R06 wanted faster loading. Fix: pre-cache moments.

Success is more users reaching the photo, and the biggest risk is that the win sits where memory is richest

North Star: 28-day URR

Goal: +5 pts URR in an 8-week user-level A/B test (~1,600 users per arm for 80% power).

No Recovery term: 0.7% of failures; tracked as a diagnostic.

Success metrics: leading & guardrail

Metric (event)	Target
Entry rate (re-entry started)	≥12% of failed searches
Clue correction (alternative taken)	15–25%
Confirm after opening a moment	≥55%
False confirmation ★	Guardrail: <3% (stop launch)
P95 time to first moment	Guardrail: <1.5 s

Risks & mitigations · Limitations below

Risk	Mitigation & evidence
R1. The win sits where memory is richest (+0.79 at L3, +0.07 at L1)	Lean on moment recognition (moment@5 0.333 at L2); measure L1 in the A/B.
R2. Date filters hide photos	Soft scoring built: L3 recall 0.851 → 0.962.
R3. Can't read text in photos (0.000 on receipts)	Add OCR text to the index.
R4. False confirmation (worst failure)	Never auto-confirm; show the surrounding photos.
R5. Ask Photos may already suffice	4 of 4 survey and 3 of 3 test users got "related photos, not the one".

Limitations: Play Store is 86.2% of posts · Hinglish is a weak signal · URR baselines modelled · ranking weights heuristic · small unmoderated samples.